

Patient-Reported Outcomes

A Value Set for EQ-5D-5L in the United Arab Emirates

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ABSTRACT

Objectives: This study aimed to develop a value set for the EQ-5D-5L based on preferences of the general adult population of the United Arab Emirates (UAE).

Methods: The study followed the EuroQol EQ-5D-5L valuation protocol and involved conducting interviewer-administered face-to-face or online interviews in Arabic or English, using the EuroQol Valuation Technology with a sample of 1005 adults representing the UAE general population. Sample recruitment involved a 2-stage quota sampling strategy across the 7 emirates of the UAE, ensuring representation of nationals and expatriates. Various models using composite time trade-off data only, discrete choice experiment data only, and hybrid using both composite time trade-off and discrete choice experiment data were examined, along with various sensitivity analyses to examine the robustness of the models.

Results: The average age of respondents was 39 years (SD 10.8), 44.5% were female, and 11% were UAE nationals. The best-performing model to generate the value set for the EQ-5D-5L was the hybrid tobit model censored at -1.0 , corrected for heteroskedasticity. Values ranged from -0.654 for the worst health state (55555) to 1 for full health (11111) and 0.962 for 11211, with 15.3% of predicted values worse than dead. Mobility problems had the largest impact on health state preference values relative to other dimensions.

Conclusion: This value set will facilitate the application and use of the EQ-5D-5L instrument in the UAE population in generating local evidence on the cost-effectiveness of healthcare interventions, as well as to enhance other applications of EQ-5D in population health assessment and health systems.

Keywords: EQ-5D-5L, quality of life, stated preferences, United Arab Emirates, utility, value.

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Highlights

- This study represents the first valuation of the EQ-5D-5L conducted in the United Arab Emirates, a country characterized by a distinctive demographic landscape, where approximately 90% of the population consists of expatriates.
- The best-performing model to generate the value set for the EQ-5D-5L was the hybrid tobit model censored at -1.0 , corrected for heteroskedasticity. Values ranged from -0.654 for the worst health state (55555) to 1 for full health (11111), with 15.3% of predicted values worse than dead. Mobility problems had the largest impact on health state preference values relative to other dimensions.
- This value set will facilitate the application and utilization of the EQ-5D-5L instrument to generate local evidence on the cost-effectiveness of healthcare interventions, aiding policy makers in making informed decisions about resource allocation and healthcare priorities.

Introduction

The ever-evolving landscape of therapeutic and medical advancements has led to a surge in healthcare technologies, accompanied by escalating healthcare costs. In response to this global phenomenon, health technology assessment (HTA) has emerged as a crucial tool, supporting evidence-based decision making to ensure both affordability and equity in healthcare resource allocation worldwide.¹ The World Health Organization recognizes HTA as pivotal to achieving universal health coverage, a tool experiencing increased adoption not only in developed nations but also in developing countries.²

Despite the absence of national HTA policies in the United Arab Emirates (UAE) at the time of writing this article, a decree from the Department of Health in Abu Dhabi marked a significant stride. The decree establishes a new entity for HTA, mandating a comprehensive value assessment process for all new health technologies and therapeutic practices in the Emirate of Abu Dhabi, including a “local (UAE) cost effectiveness analysis” (Circular No. 31/2021). Anticipating a need for local evidence-based data, this decree is a precursor for similar initiatives in other emirates to rationalize healthcare expenditure and

resource allocation and underscores the need to establish infrastructure supporting robust and localized HTA practices in the UAE.³

EQ-5D stands as one of the most widely used measures of health-related quality of life worldwide. Its application in generating quality-adjusted life-years for economic evaluations has made it a cornerstone in these domains.^{4–6} However, the absence of a country-specific value set based on the UAE general population preferences hinders its full potential use in the local context.^{7–12} In addition, there is a growing movement in the UAE toward value-based care models and the use of patient-reported outcome measures (PROMs) as an approach in the comprehensive assessment of value.¹³

Within the UAE’s research landscape, there is a notable gap in the application of EQ-5D. Although previous studies have demonstrated the feasibility of conducting valuation work for the EQ-5D, no value set currently exists for any of its instruments.¹⁴

This study aimed to address this gap by developing a value set for the EQ-5D-5L based on the preferences of the UAE general population. The resulting value set will enable economic evaluation studies to incorporate local preferences, thereby enhancing their relevance to local decision-making processes. Furthermore, this research supports the broader initiative in the UAE to integrate patients' perspectives into health outcome measurement, positioning the EQ-5D as a valuable PROM in the country's healthcare landscape.¹⁰

Methods

Setting

The UAE consists of 7 emirates: Abu Dhabi, Sharjah, Dubai, Fujairah, Ras Al Khaimah, Ajman, and Umm Al Quwain, with an estimated population of 10.2 million.¹⁵ The demographic landscape is distinctive, with only 11.8% of the population being Emirati nationals, whereas 88.2% are expatriates.¹⁶ Local stakeholders emphasized the importance of including UAE nationals and expatriates in this study, given that decisions affecting healthcare services or technologies will affect both citizens and noncitizens. Although the expatriate population is often transient, many have lived their entire lives in the UAE, making their perspectives relevant despite being ineligible for citizenship. The literacy rate in the UAE stands at 98%, with Arabic as the official language and English recognized as the second official language.¹⁵

Study Design

This study followed the EuroQol valuation protocol for the EQ-5D-5L.¹⁷⁻¹⁹ Using the EuroQol Valuation Technology (EQ-VT) platform, we conducted a computer-based, interviewer-administered face-to-face or online survey with a representative sample of the general UAE population. Following the reporting standards for valuation studies of multiattribute utility instruments, as outlined in the Checklist for Reporting Valuation Studies of Multi-Attribute Utility-Based Instruments checklist,²⁰ we present our findings in this article. Ethical approval for the study was granted by the Social Sciences Research Ethics Committee at the United Arab Emirates University (ERSC_2022_1648), and written informed consent was obtained from each participant.

Descriptive System

The EQ-5D-5L includes a descriptive system and the EuroQol visual analog scale (EQ-VAS). The descriptive system assesses 5 dimensions of health: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression (Appendix Fig 1). Each dimension has 5 levels of severity—no, mild, moderate, severe, and extreme/unable to—describing 3125 distinct health states.²¹ Represented by a 5-digit number, these health states range from 11111 (indicating no problems across all dimensions) to 55555 (signifying extreme problems or inability in every dimension). The EQ-VAS captures the patient's self-rated health on a scale ranging from 0 "the worst health you can imagine" to 100 "the best health you can imagine."

Preference Elicitation Methods and Health State Selection

The EuroQol EQ-5D-5L valuation protocol uses 2 preference elicitation methods: composite time trade-off (cTTO) and discrete choice experiment (DCE). A detailed description of these methods and the valuation design has been previously reported.¹⁷⁻¹⁹ Briefly, the cTTO method evaluates an EQ-5D-5L health state in relation to full health. It uses conventional time trade-off (TTO) to determine

values for health states perceived as better than dead and lead-time TTO for states viewed as worse than dead (Appendix Fig. 2 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.01.003>). After completing the cTTO task, respondents see a ranked list of the 10 valued health states based on their TTO scores (the feedback module), with the highest values at the top. Health states that receive the same score are displayed side by side. Participants can review this ranking and indicate whether they would like to change the order of any health state. Any flagged states are reviewed as part of the quality control (QC) process to assess their impact on valuation inconsistencies. Although participants cannot alter the valuation of a health state, they can flag concerns in the feedback module. Afterward, respondents complete the DCE tasks where they express preferences between 2 different EQ-5D-5L health state profiles in pairwise comparisons (Appendix Fig. 3 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.01.003>).

Sampling and Recruitment Strategy

A multicounty pilot study that informed various aspects of the EQ-5D-5L valuation protocol concluded that a minimum sample size of 1000 respondents per country is recommended for effective value set modeling.^{17,18} Accordingly, 1150 adult UAE participants were targeted, with 150 participants added to the minimum sample to account for possible attrition due to data quality issues. Recruitment used a 2-stage quota sampling strategy. First, we stratified geographically by Emirate, and then strata were defined based on age and sex distribution of the general population¹⁶ (Appendix Table 1 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.01.003>).

Adults aged 18 years and older who meet specific criteria within defined sampling quotas, ie, either Emirati nationals or non-nationals who have resided in the UAE for at least 5 years and can complete the interview in Arabic or English, were included. The 5-year requirement was established to ensure that participants have sufficient lived experience in the UAE, allowing their preferences to reflect the local context rather than their countries of origin. This timeframe was determined based on consultations with local stakeholders who agreed it was adequate for capturing relevant perspectives.

Recruitment was done through various approaches and spaces to ensure the wide representativeness of the general population. Strategies included leveraging the personal and professional networks of study interviewers and project team members, using snowball sampling, and engaging individuals in public spaces such as community centers, university campuses, government buildings, shopping malls, parks, and sports facilities. Study recruitment posters were displayed in these locations and shared across various social media platforms. To further enhance demographic representation, recruitment efforts were scheduled at different times and on various days of the week.

Data Collection

Practice interviews were conducted between November and December 2022, and main data collection between January and August 2023. Upon recruitment, participants were offered the option of completing the interview in person or online. In-person interviews were conducted at the Center for Clinical, Health Economics and Outcomes Research office; university campus facilities; participant's workplace; or a public space depending on the participant's preference. Participants received a 70 UAE dirhams (equivalent to 17.5 Euros) gift card for their participation in the study.

The interview process consisted of the following steps: (1) interviewer explained the purpose of the study, its procedure, confidentiality, and risk/benefits to the participant; (2) participant completed the EQ-5D-5L; (3) interviewer explained (using the wheelchair example) and administered the cTTO task (10 health states); (4) interviewer explained and administered the DCE task (7 pairs); and (5) participant completed a country-specific questionnaire including questions on sociodemographics, health history, and feedback on TTO and DCE tasks.

All study materials were translated into Arabic. The EQ-5D-5L Arabic version for UAE was used in this study. For English-speaking participants, the English EQ-5D-5L version validated for the United Kingdom was used as no English-specific version for the UAE is available.

Interviewer Training and QC Process

Recruited interviewers ($n = 14$) received a rigorous 2-day training that included (1) an introduction to measuring and valuing health using the EQ-5D-5L, (2) explanation of the EQ-VT protocol and interviewer instructions, (3) quality criteria and QC process, (4) practice in groups, and (5) pretesting of the tools in selected population. In addition, an online 3-hour review session was provided to interviewers after completing 2 practice interviews each.

Before the main data collection started, each interviewer completed 5 practice interviews.²² Data quality was reviewed by the study lead and 2 QC officers from the EuroQol EQ-VT support team, and extensive individual feedback was provided to the interviewers. The QC process focused on checking compliance with the protocol, evaluating the interviewer effect, and assessing face validity of the collected data.²²

Three rounds of these practice interviews and QC were completed until the data quality was considered satisfactory to allow the interviewer to start data collection. After this process, 2 interviewers dropped out due to personal reasons, and 5 were not allowed to proceed with data collection due to significant concerns about data quality. Four of the 7 retained interviewers were fluent in Arabic and English; all of them had graduate-level education, with medical or health research background.

QC review and discussions were held between the study team and the EQ-VT support team on a weekly basis in the first 4 months and biweekly thereafter. Data collection was paused after completing 25%, 50%, and 75% of interviews for an evaluation of the quality of collected data and to conduct interim analysis.

Given that the valuation tasks were conducted in both English and Arabic and through in-person and online formats, we compared data quality across these subgroups during each QC check. The differences in all quality parameters were minimal. Therefore, data from both languages and collection methods were combined to generate the value set.

Data Analysis and Model Selection

Descriptive statistics were used to examine sample characteristics and responses to the cTTO and DCE. Statistical modeling was used to estimate the values for all EQ-5D-5L health states. We examined a total of 10 models:

1. random intercept model for cTTO data
2. random intercept tobit model, with censoring at -1 , for cTTO data
3. model correcting for heteroskedasticity for cTTO data
4. model correcting for heteroskedasticity without an intercept for cTTO data

5. tobit model correcting for heteroskedasticity without an intercept for cTTO data
6. conditional logit model for DCE data
7. hybrid model combining cTTO and DCE data
8. hybrid model with a tobit link for cTTO data, with censoring at -1
9. hybrid model correcting for heteroskedasticity in cTTO data
10. hybrid model correcting for heteroskedasticity and using a tobit link with censoring at -1 for cTTO data

In these regression models, EQ-5D-5L index and cTTO values served as dependent variables. The independent variables included 20 dummy variables representing each dimension level from mobility level 2 to anxiety/depression level 5, using level 1 as the reference. Models 1, 2, 3, and 5 estimated 21 main effect parameters and a constant, whereas the others did not include a constant. The hybrid models also estimated a scaling parameter (θ) to align values for cTTO and DCE data. Detailed model specifications can be found in Bailey et al.²³

Sensitivity analyses involved 4 sets of models:

1. including all observations
2. excluding those with inconsistencies for health state 55555
3. excluding valuations flagged in the feedback module
4. excluding both the previous groups

Model performance was assessed by evaluating the logical consistency of parameter estimates, model parsimony, and various goodness-of-fit statistics, including the Akaike information criterion (AIC), Bayesian information criterion (BIC), root mean square error (RMSE), and mean absolute error (MAE) for the means of the 86 states in the cTTO design.^{24,25} AIC and BIC were specifically used to compare cTTO-only models with each other and hybrid models with each other. MAE and RMSE were used to compare among tobit models and among nontobit models. When model performance was comparable, hybrid models using all collected data were preferred. In addition, tobit models were favored due to their ability to account for the censored nature of the data.

We also conducted cross-validation to evaluate the out-of-sample predictivity of each model. This involved removing 1 health state from the 86-state design, estimating each model on the remaining 85 states, and predicting the value for the excluded state. This process was repeated for all 86 states. We calculated MAE and RMSE for these predictions. A similar analysis was conducted by removing blocks of health states; 1 block of 10 states was omitted, and models were estimated on the remaining 9 blocks, predicting values for the excluded block. This was repeated for each block, and MAE and RMSE were computed across all 10 blocks. All models were estimated using Stata 18.0 (StataCorp LLC, College Station, TX).

Results

General Characteristics of the Sample

A total of 1145 interviews were conducted, and data from 1005 respondents were included in the analysis, providing a total of 10 050 cTTO responses and 7035 DCE responses. The 140 excluded interviews comprised practice interviews conducted in the first 6 weeks as part of the QC process. There were negligible differences in characteristics between the study sample and those excluded.

The mean age of respondents was 32.1 years (SD 11.4; range 18–76 years), 44.4% were female, and 11.4% were UAE nationals (Table 1). Although most participants were expatriates, this distribution reflects the overall national and expatriate demographics

Table 1. General characteristics of the sample.

Characteristic	Study sample (N = 1005)	Excluded sample (n = 140)	UAE general population
Age group (years), n (%)			
18-39	732 (72.8)	109 (77.9)	65.9
40-64	265 (26.4)	31 (22.1)	32.7
65 or older	8 (0.8)	0 (0.0)	1.4
Sex, n (%)			
Male	559 (55.62)	75 (53.6)	69.2
Female	446 (44.38)	65 (46.4)	30.8
Citizenship status			
National	115 (11.4)	2 (1.4)	11.8
Expat	890 (88.6)	138 (98.6)	88.2
Emirate of residence			
Dubai	361 (35.9)	55 (39.2)	43.8
Abu Dhabi	112 (11.1)	19 (13.6)	19.3
Sharjah	332 (33.0)	38 (27.2)	22.4
Ajman	160 (15.9)	27 (19.3)	6.3
Ras Al Khaimah	13 (1.3)	0 (0.0)	4.3
Umm Al Quwain	20 (2.0)	1 (0.7)	0.6
Fujairah	8 (0.8)	0 (0.0)	3.2
Marital status			Not available
Single	468 (46.6)	59 (42.1)	
Married	506 (50.3)	77 (55.0)	
Divorced/separated	11 (1.1)	1 (0.7)	
Widowed	4 (0.4)	0 (0.0)	
Prefer not to respond	16 (1.6)	3 (2.1)	
Educational level			Not available
Less than high school	18 (1.8)	3 (2.1)	
Completed high school	338 (33.6)	41 (29.3)	
Bachelor's degree	387 (38.5)	56 (40.0)	
Post graduate	145 (14.4)	18 (12.9)	
Technical school	62 (6.2)	9 (6.4)	
Prefer not to respond	12 (1.2)	4 (2.9)	
Other	43 (4.3)	9 (6.4)	
Monthly household income			Not available
Less than AED5000	155 (15.5)	11 (8.1)	
AED5000-AED9999	160 (15.9)	23 (16.7)	
AED10 000-AED24 999	197 (19.6)	34 (24.1)	
AED25 000-AED45 999	87 (8.7)	7 (4.9)	
AED50 000 or more	25 (2.5)	6 (4.3)	
Prefer not to respond	380 (37.8)	59 (41.9)	
Religion			Not available
Muslim	803 (79.9)	115 (82.1)	
Christian	94 (9.4)	10 (7.1)	
Hindu	98 (9.7)	7 (5.0)	
Buddhist	3 (0.3)	0 (0.0)	
Other	4 (0.4)	0 (0.0)	
Prefer not to respond	3 (0.3)	8 (5.7)	

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Table 1. Continued

Characteristic	Study sample (N = 1005)	Excluded sample (n = 140)	UAE general population
EQ-5D-5L			Not available
Mobility			
None	905 (90.1)	132 (94.3)	
Slight	79 (7.9)	6 (4.3)	
Moderate	15 (1.5)	2 (1.4)	
Severe	4 (0.4)	0 (0.0)	
Unable/extreme	2 (0.2)	0 (0.0)	
Self-care			
None	979 (97.4)	138 (98.6)	
Slight	20 (2.0)	2 (1.4)	
Moderate	4 (0.4)	0 (0.0)	
Severe	1 (0.1)	0 (0.0)	
Unable/extreme	1 (0.1)	0 (0.0)	
Usual activities			
None	870 (86.6)	128 (91.4)	
Slight	116 (11.5)	11 (7.9)	
Moderate	16 (1.6)	1 (0.7)	
Severe	0 (0.0)	0 (0.0)	
Unable/extreme	3 (0.3)	0 (0.0)	
Pain/discomfort			
None	658 (65.5)	82 (58.6)	
Slight	292 (29.1)	49 (35.0)	
Moderate	53 (5.27)	9 (6.4)	
Severe	2 (0.2)	0 (0.0)	
Unable/extreme	0 (0.0)	0 (0.0)	
Anxiety/depression			
None	678 (67.5)	94 (67.2)	
Slight	245 (24.4)	36 (25.7)	
Moderate	70 (6.9)	10 (7.1)	
Severe	9 (0.9)	0 (0.0)	
Unable/extreme	3 (0.3)	0 (0.0)	
EQ-VAS	86.0 ± 11.4	84.9 ± 13.7	

AED indicates UAE dirhams; EQ-VAS, EuroQol visual analog scale.

in the UAE. Among expatriates, the average number of years lived in the UAE was 15.6 years (SD 8.5; range 5–46). Notably, 63.1% planned to remain in the UAE for the next 10 years, 31.9% were uncertain, and only 5% indicated they would not stay.

Most respondents (79.9%) identified as Muslim, with more than two-thirds (69.2%) who reported that their religious beliefs did not influence their responses during the valuation task.

Overall, the sample was representative of the adult general population in the UAE regarding age, sex, and geographic distribution. Although no published estimates exist for the distribution of marital status, educational attainment, and household income in the UAE general population, this sample adequately represented all subgroups based on these parameters.

Regarding health status as measured by the EQ-5D-5L, 9.9% reported mild to severe problems in mobility, 2.6% in self-care, 13.4% in usual activities, 34.5% in pain/discomfort, and 32.5% in anxiety/depression. The mean EQ-VAS score was 86.0 (SD 11.4).

cTTO and DCE Data

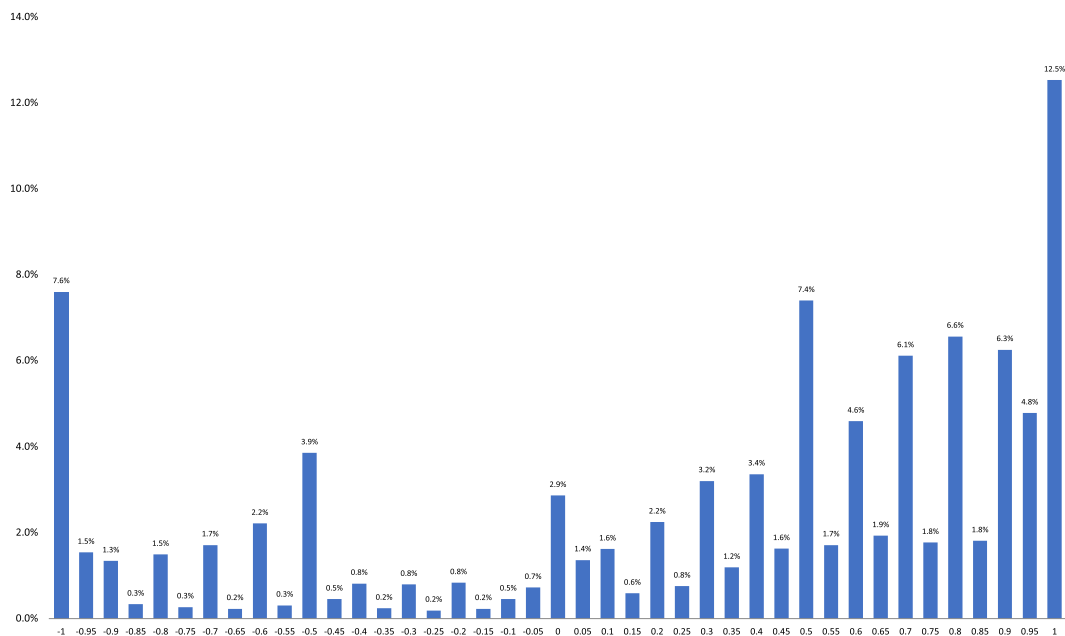
The mean iterative steps to reach the point of indifference was 8.6 (SD 5.6). The mean time spent in the feedback module was 2.3 minutes (SD 1.7). Participants flagged 1018 cTTO responses (10.1%) using the feedback module. A total of 285 participants (28.4%) had at least one inconsistency, which reduced to 160 (15.9%) after using the feedback module. Stated preferences or valuations for different health states do not follow a logically expected or hierarchical order; eg, a more severe health state is assigned a higher

value, indicating better health, than a less severe health state; in the case of health state 55555, an inconsistency arises when a respondent assigns it a higher value by at least 0.5 points than any other health state that is objectively and logically less severe. The number of inconsistencies related to severity “6” (mild problem in one dimension only) and 55555 states was 25 (2.5%) and 84 (8.4%), respectively, and reduced to 8 (0.8%) and 24 (2.4%), respectively, after using the feedback module.

The main analysis included all cTTO valuations; 25.7% of these were considered WTD, and the mean observed value was negative for 11 of the 86 health states included in the cTTO design. The percentages of values clustered at −1.0, −0.5, 0, 0.5, and 1.0 were 7.6%, 3.9%, 2.9%, 7.4%, and 12.5%, respectively (Fig. 1). Given that the level sum score increased for the EQ-5D-5L health states, lower mean TTO values and a larger SD were observed (Fig. 2). The mean (SD) observed cTTO value of the 86 health states was 0.303 (SD 0.657), which ranged from −0.538 (SD 0.510) for the worst health state (55555) to 0.948 (SD 0.116) for the very mild health states (21111, 12111, 11211, 11121, 11112).

Modeling Results

Modeling results for all 10 models are presented in Table 2. All models exhibited logical consistency, with a single inconsistency identified in the mobility dimension level 3 within the heteroskedastic tobit model for cTTO data. Most parameter estimates were statistically significant, except for 1 parameter related to pain/discomfort dimension level 3 in the same model. The

Figure 1. Distribution of observed composite time trade-off (cTTO) values.

agreement between predicted values in the cTTO heteroskedastic tobit model and those in the DCE conditional logit model was significantly high (Fig. 3).

In terms of dimension importance, mobility was the most critical dimension in 7 of the 10 models, whereas pain/discomfort ranked

highest in the 3 heteroskedastic models using only cTTO data. Anxiety/depression consistently ranked third in 7 models, followed by self-care and usual activities in fourth and fifth places, respectively.

Cross-validation results showed comparable MAE and RMSE values between the tobit models (comparing both cTTO-only tobit

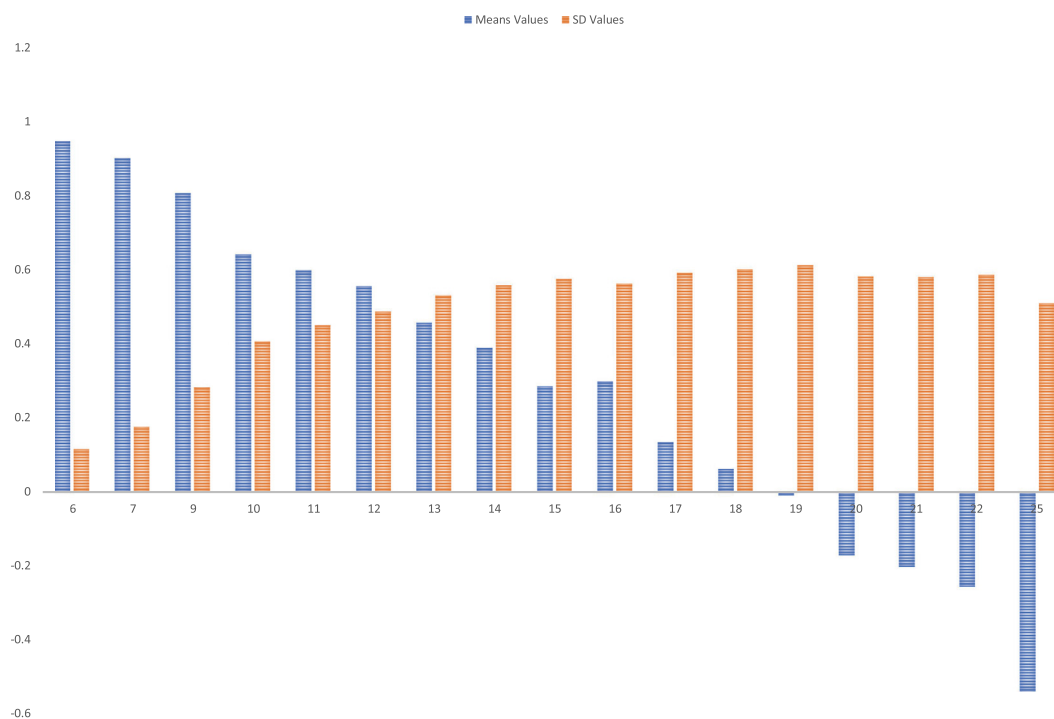
Figure 2. Mean observed composite time trade-off (cTTO) values by level sum scores.

Table 2. Parameter estimates and model fit statistics for all models.

Models	Random intercept cTTO only			Random intercept tobit cTTO only			Heteroskedastic cTTO only			Heteroskedastic (NC) cTTO only			Heteroskedastic tobit cTTO only		
Coefficients	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value
mo2	0.065	0.013	.000	0.061	0.014	.000	0.057	0.012	.000	0.060	0.010	.000	0.053	0.012	.000
mo3	0.068	0.014	.000	0.060	0.015	.000	0.061	0.017	.000	0.062	0.017	.000	0.045	0.018	.013
mo4	0.241	0.015	.000	0.238	0.016	.000	0.251	0.019	.000	0.252	0.019	.000	0.250	0.020	.000
mo5	0.364	0.014	.000	0.379	0.015	.000	0.351	0.016	.000	0.352	0.016	.000	0.378	0.017	.000
sc2	0.054	0.013	.000	0.054	0.014	.000	0.054	0.011	.000	0.056	0.009	.000	0.048	0.011	.000
sc3	0.092	0.015	.000	0.092	0.016	.000	0.100	0.016	.000	0.099	0.016	.000	0.094	0.017	.000
sc4	0.232	0.015	.000	0.238	0.016	.000	0.222	0.018	.000	0.223	0.017	.000	0.229	0.019	.000
sc5	0.276	0.014	.000	0.297	0.015	.000	0.275	0.015	.000	0.275	0.015	.000	0.300	0.017	.000
ua2	0.047	0.014	.001	0.047	0.015	.001	0.043	0.011	.000	0.046	0.009	.000	0.041	0.011	.000
ua3	0.075	0.015	.000	0.074	0.016	.000	0.088	0.016	.000	0.089	0.016	.000	0.088	0.017	.000
ua4	0.191	0.015	.000	0.192	0.016	.000	0.181	0.017	.000	0.183	0.016	.000	0.178	0.017	.000
ua5	0.208	0.013	.000	0.225	0.014	.000	0.195	0.016	.000	0.196	0.016	.000	0.219	0.017	.000
pd2	0.042	0.013	.001	0.041	0.013	.002	0.036	0.010	.000	0.038	0.008	.000	0.035	0.010	.000
pd3	0.056	0.015	.000	0.056	0.016	.001	0.044	0.018	.016	0.043	0.018	.017	0.035	0.019	.063
pd4	0.253	0.013	.000	0.263	0.014	.000	0.257	0.016	.000	0.257	0.016	.000	0.268	0.017	.000
pd5	0.343	0.014	.000	0.359	0.015	.000	0.377	0.018	.000	0.379	0.017	.000	0.399	0.019	.000
ad2	0.050	0.015	.001	0.047	0.015	.002	0.058	0.010	.000	0.060	0.009	.000	0.057	0.010	.000
ad3	0.112	0.016	.000	0.107	0.018	.000	0.120	0.018	.000	0.121	0.017	.000	0.113	0.018	.000
ad4	0.283	0.015	.000	0.284	0.016	.000	0.290	0.016	.000	0.291	0.016	.000	0.295	0.017	.000
ad5	0.341	0.014	.000	0.356	0.015	.000	0.343	0.015	.000	0.343	0.015	.000	0.366	0.016	.000
Constant	0.006	0.018	.749	0.005	0.019	.785	0.004	0.010	.684	N/A	N/A	N/A	N/A	N/A	N/A
Dimension ranking	MO-PD-AD-SC-UA			MO-PD-AD-SC-UA			PD-MO-AD-SC-UA			PD-MO-AD-SC-UA			PD-MO-AD-SC-UA		
AIC	11 361.03			13 306.92			12 905.15			12 903.31			14 464.28		
BIC	11 526.98			13 472.87			13 208.19			13 199.14			14 767.32		
RMSE (86 states)	0.055			0.061			0.056			0.055			0.067		
MAE (86 states)	0.040			0.044			0.039			0.039			0.045		
Value (55555)	−0.538			−0.621			−0.545			−0.544			−0.666		

Coefficients	DCE conditional logit			Hybrid			Hybrid tobit			Hybrid Heteroskedastic			Hybrid Heteroskedastic tobit (preferred model)		
	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value
mo2	0.478	0.057	.000	0.097	0.009	.000	0.097	0.010	.000	0.079	0.007	.000	0.075	0.007	.000
mo3	0.611	0.068	.000	0.114	0.011	.000	0.115	0.011	.000	0.102	0.010	.000	0.100	0.011	.000
mo4	1.266	0.069	.000	0.263	0.010	.000	0.268	0.011	.000	0.254	0.010	.000	0.261	0.011	.000
mo5	2.173	0.085	.000	0.420	0.010	.000	0.440	0.011	.000	0.412	0.010	.000	0.438	0.011	.000
sc2	0.371	0.061	.000	0.066	0.009	.000	0.065	0.010	.000	0.061	0.007	.000	0.057	0.007	.000
sc3	0.380	0.068	.000	0.080	0.011	.000	0.080	0.011	.000	0.083	0.010	.000	0.082	0.011	.000
sc4	0.900	0.070	.000	0.196	0.010	.000	0.199	0.011	.000	0.191	0.010	.000	0.196	0.011	.000
sc5	1.190	0.067	.000	0.243	0.010	.000	0.255	0.010	.000	0.252	0.009	.000	0.268	0.010	.000
ua2	0.220	0.059	.000	0.036	0.009	.000	0.034	0.010	.000	0.039	0.007	.000	0.038	0.007	.000
ua3	0.266	0.066	.000	0.059	0.010	.000	0.057	0.011	.000	0.064	0.010	.000	0.062	0.010	.000
ua4	0.703	0.069	.000	0.152	0.010	.000	0.153	0.011	.000	0.156	0.010	.000	0.159	0.010	.000
ua5	1.083	0.071	.000	0.219	0.010	.000	0.229	0.010	.000	0.218	0.009	.000	0.232	0.010	.000
pd2	0.340	0.062	.000	0.057	0.009	.000	0.057	0.010	.000	0.048	0.007	.000	0.047	0.007	.000
pd3	0.431	0.067	.000	0.074	0.011	.000	0.074	0.011	.000	0.074	0.010	.000	0.072	0.011	.000

continued on next page

Table 2. Continued

Coefficients	DCE conditional logit			Hybrid			Hybrid tobit			Hybrid Heteroskedastic			Hybrid Heteroskedastic tobit (preferred model)		
	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value	Beta	SE	P value
pd4	1.193	0.071	.000	0.246	0.010	.000	0.254	0.011	.000	0.249	0.010	.000	0.261	0.011	.000
pd5	1.562	0.073	.000	0.324	0.010	.000	0.340	0.011	.000	0.336	0.010	.000	0.359	0.011	.000
ad2	0.350	0.065	.000	0.068	0.010	.000	0.066	0.010	.000	0.063	0.007	.000	0.062	0.007	.000
ad3	0.624	0.068	.000	0.126	0.011	.000	0.125	0.011	.000	0.128	0.010	.000	0.129	0.011	.000
ad4	1.197	0.073	.000	0.263	0.010	.000	0.267	0.010	.000	0.261	0.010	.000	0.268	0.010	.000
ad5	1.640	0.076	.000	0.334	0.010	.000	0.347	0.010	.000	0.336	0.009	.000	0.357	0.010	.000
Constant	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Dimension ranking	MO-AD-PD-SC-UA			MO-AD-PD-SC-UA			MO-AD-PD-SC-UA			MO-PD-AD-SC-UA			MO-PD-AD-SC-UA		
AIC	7299.84			22 100.39			24 167.26			20 233.45			21 794.74		
BIC	7437.01			22 270.80			24 337.67			20 558.78			22 120.07		
RMSE (86 states)	N/A			0.062			0.067			0.062			0.070		
MAE (86 states)	N/A			0.045			0.051			0.045			0.052		
Value (55555)	7.648			−0.541			−0.611			−0.555			−0.654		

AD indicates anxiety/depression; AIC, Akaike information criterion; BIC, Bayesian information criterion; cTTO, composite time trade-off; DCE, discrete choice experiment; MAE, mean absolute error; MO, mobility; N/A, not available; NC, no constant; PD, pain/discomfort; RMSE, root mean square error; SC, self-care; SE, standard error; UA, usual activities.

models and hybrid tobit models, with or without correcting for heteroskedasticity), as well as between nontobit models (eg, random intercept and heteroskedastic cTTO models, as well as hybrid and hybrid heteroskedastic models). This similarity in out-of-sample predictivity suggested robustness across model types, regardless of whether individual states or blocks of states were omitted.

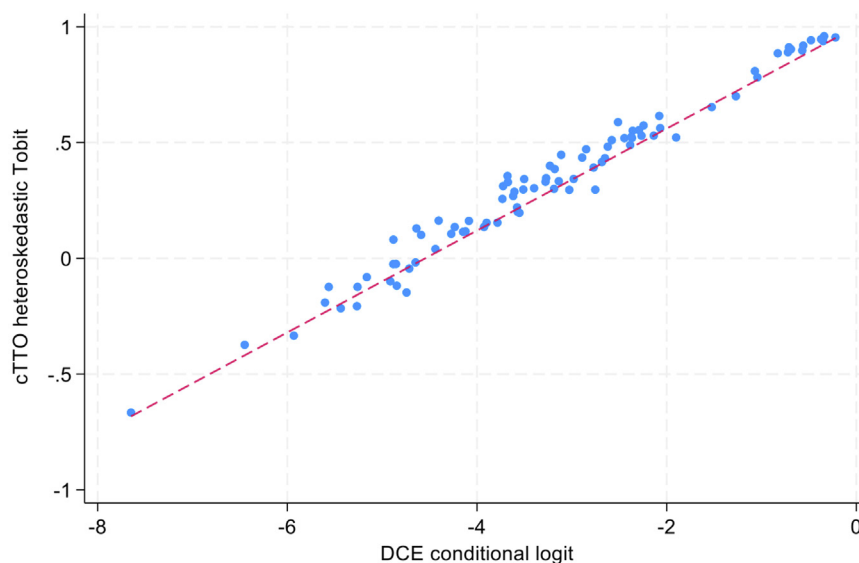
Sensitivity analysis indicated that results remained consistent in terms of parameter estimates and model fit statistics compared

with the complete data set (data not indicated; available upon request). Consequently, we opted to report results without excluding any observations.

Preferred Model and Value Set

The hybrid heteroskedastic model and the hybrid heteroskedastic tobit model outperformed their counterparts in prediction accuracy, logical consistency, significance level, and goodness of fit. Among these, the hybrid tobit model, censored at −1.0 and

Figure 3. Agreement between predicted values in the cTTO heteroskedastic tobit and DCE conditional logit models.



cTTO indicates composite time trade-off; DCE, discrete choice experiment.

Table 3. An EQ-5D-5L value set for UAE.

	Coefficient	Value for health state 32234
Constant	1.000	
Mobility		
Level 2: slight	0.075	0.100
Level 3: moderate	0.100	
Level 4: severe	0.261	
Level 5: unable	0.438	
Self-care		
Level 2: slight	0.057	0.057
Level 3: moderate	0.082	
Level 4: severe	0.196	
Level 5: unable	0.268	
Usual activities		
Level 2: slight	0.038	0.038
Level 3: moderate	0.062	
Level 4: severe	0.159	
Level 5: unable	0.232	
Pain/discomfort		
Level 2: slight	0.047	0.072
Level 3: moderate	0.072	
Level 4: severe	0.261	
Level 5: extreme	0.359	
Anxiety/depression		
Level 2: slight	0.062	0.268
Level 3: moderate	0.129	
Level 4: severe	0.268	
Level 5: extreme	0.357	
The value for health state 32234	$1 - (0.100 + 0.057 + 0.038 + 0.072 + 0.268)$	= 0.465

UAE indicates United Arab Emirates.

accounting for heteroskedasticity, was deemed the preferred model for representing the value set for the UAE (Table 3). This model effectively used all available data, addressed the censored nature of the cTTO data, and demonstrated comparable RMSE and MAE with other tobit models while outperforming them in AIC and BIC metrics.

The predicted cTTO values ranged from − 0.654 for the worst health state (55555) to 0.962 for the health state 11211. The ranking of dimensions by relative importance was as follows: mobility (most important), pain/discomfort, anxiety/depression (similar to pain/discomfort), self-care, and usual activities (least important). Utility values for any health state can be calculated by subtracting the parameter estimates for each dimension level from 1 (Table 3).

Discussion

This EQ-5D-5L valuation study aimed to generate a local value set that reflects how the general adult population in the UAE values different health states and understands which aspects of health matter most to this population. Having a local value set for the EQ-5D-5L will allow calculating quality-adjusted life-years for use in local cost-utility analysis, thereby generating essential and relevant evidence for the UAE healthcare system. In addition, the initiative will help leverage the diverse applications of the EQ-5D

in clinical studies and population health assessment and as a PROM in health systems in the UAE.

In 2015, a feasibility study explored the use and cultural appropriateness of the EQ-5D valuation methods in the UAE and found these methods to be useful with some adaptation to the local cultural context, such as increasing the emphasis in the instructions that the objectives of the study are not intended to cause conflict with respondents' religious beliefs.^{14,26} Since that study was conducted, several valuation studies aiming to generate value sets for the 3- or 5-level versions of the EQ-5D have been successfully conducted in countries with similar religious and cultural profiles including Pakistan,²⁷ Egypt,²⁸ Indonesia,²⁹ Saudi Arabia,³⁰ and Tunisia.³¹

The UAE general population reported fewer problems across all EQ-5D-5L dimensions compared with many other populations,^{28,30-37} which is likely due to the youthful nature of its population. Nonetheless, it is worth noting that 34.5% and 32.5% of the participants reported mild to severe problems in the pain/discomfort and anxiety/depression dimensions, respectively. This is consistent with previous reports of the high prevalence of anxiety and depression disorders in the general population in the UAE.³⁸ The finding with regard to the pain/discomfort dimension warrants further attention from a public health perspective, given that the increasing prevalence and incidence of chronic pain have always been considered a major public health concern and a leading cause of disability worldwide.³⁹⁻⁴¹

Problems in mobility were ranked the highest relative to other EQ-5D-5L dimensions, a finding similar to that in the Egyptian valuation study of the EQ-5D-5L,²⁸ but different from its neighboring country Saudi Arabia where pain/discomfort was ranked the highest.³⁰ The predicted value of the worst health states (55555) of −0.654 was very close to that in Saudi Arabia (−0.683),³⁰ higher than the value reported in Egypt (−0.964),²⁸ but lower than that in the United States (−0.573).³⁵

Conducting this valuation study in the UAE provided valuable insights into a unique demographic landscape, consisting of 11.8% of UAE nationals and 88.2% of expatriates. This population is characterized by its transient nature, resulting in a constantly evolving composition. Notably, no previous valuation studies of the EQ-5D or other preference-based measures have been conducted in a country with a similar demographic profile, making direct comparisons difficult. Although we did not initially set out to analyze how expatriate status influenced health state preferences or decision making during the valuation task, many participants indicated that their expatriate status affected their choices. Given that their residency in the UAE is tied to employment, any health state affecting their job performance led them to value life-years differently. This observation was corroborated by numerous testimonials from interviewers and field notes.

Although the data collected in this study do not allow for a comprehensive examination of expatriate status on health state valuations, future qualitative research could explore this further. It is essential to recognize the significant variability in expatriates' lived experiences, which include factors such as years lived in the UAE (with some individuals born and raised there), education level, employment type, income, language, family structure, country of origin, health insurance, access to services, and eligibility for long-term visas.

The UAE's demographic composition raises important questions about what constitutes the general population and how to define a representative sample in such a context. As many countries experience increasing numbers of immigrants and refugees, along with ongoing globalization, the composition of the general population is continually shifting. This dynamic also raises

concerns about the time validity of value sets and the frequency with which they should be updated—a discussion that lies beyond the scope of this article.

Although both Arabic and English languages were used in this study, which may lead to a potential dual language effect, it is important to recognize that previous valuation studies, such as those conducted in Canada (French and English), the United States (Spanish and English), and India (in multiple languages), have navigated similar challenges. When interpreting valuation results across different languages, we must exercise caution, given that language can often serve as a marker for distinct life experiences. For example, in Canada, most French speakers reside in Quebec and Ontario. Therefore, any observed differences in valuations may not solely stem from language but could also be influenced by geographic location and associated life experiences. A similar situation exists in the United States when comparing Spanish and English speakers. In the context of the UAE, we encounter comparable challenges. Although UAE nationals primarily speak Arabic, many expatriates also speak Arabic because they come from neighboring Arab countries. Conversely, a significant number of expatriates communicate in English, which may or may not be their native language. Consequently, differences in valuations by language could be misleading; the underlying factors may relate more to an individual's residency status and their lived experiences in the UAE rather than the language itself. To address these complexities, local stakeholders emphasized the importance of ensuring that our study sample is representative of the UAE's general population in terms of national or expatriate status, language spoken, age, gender, and socioeconomic status. This careful consideration helps enhance the relevance of our findings for HTA within the UAE context.

This study has 2 key strengths. First, it achieved a representative sample of the general UAE population in terms of age, sex, distribution across the 7 emirates, and the proportional representation of nationals and expatriates. Second, both cTTO and DCE data were of high quality, whereby excluding the small number of inconsistent observations and/or those flagged in the feedback module had an insignificant impact on the models. Notably, the study benefited from the high-quality data gathered through online interviews, which proved comparable with in-person interviews. The experience underscores the validity of online interviews as a viable method for conducting valuation studies, which was demonstrated in previous studies,^{42–44} offering distinct advantages such as broader geographic coverage within a country and also the potential to enhance representativeness.

The study has a few limitations that should be acknowledged. First, for English-speaking participants in the UAE, the EQ-5D-5L version validated for the United Kingdom was used for self-reporting and to derive the health state descriptions used in the valuation task because there was no English version of the EQ-5D-5L for the UAE population. However, restricting participation to only Arabic-speaking individuals would have significantly constrained the sample representativeness, given that a substantial proportion of non-nationals in the UAE do not speak Arabic, despite comprising a significant portion of the general population. Although using the United Kingdom's English version of EQ-5D-5L may influence direct responses to the instrument, using valuation tasks in multiple languages has been practiced in numerous countries with diverse linguistic landscapes.^{35,37,45} This raises an important question about the need for universally applicable language versions of the EQ-5D to enhance its relevance in similar settings.

Second, the inclusion criteria encompassed expatriates who had resided in the emirates for at least 5 years, under the assumption that these individuals would possess adequate

interaction and experience within the local context to value health states accurately in the valuation tasks. However, in the absence of conclusive evidence supporting this criterion, uncertainty remains regarding its sufficiency. Although a substantial majority of respondents (95%) indicated a likelihood of continuing to reside in the UAE for the next decade, it is essential to note that there is a lack of valuation studies in countries with a similar and unique population mixture of nationals and expatriates.

Conclusions

The development of an EQ-5D-5L value set for the UAE marks an important milestone in the advancement of health outcomes research and decision making in the country. This value set will facilitate the application and utilization of the EQ-5D-5L instrument in enabling researchers to generate local evidence on the cost-effectiveness of healthcare interventions, aiding policy makers in making informed decisions about resource allocation and healthcare priorities.

Author Disclosures

Author disclosure forms can be accessed below in the Supplemental Material section.

Views expressed in the paper are those of the authors and are not necessarily those of their institutions or the EuroQol Research Foundation.

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REFERENCES

- O'Rourke B, Oortwijn W, Schuller T. Announcing the new definition of health technology assessment. *Value Health*. 2020;23(6):824–825.
- World Health Organization. Developing an approach for using health technology assessments in reimbursement systems for medical products. <https://www.who.int/publications/i/item/WHO-EMP-PAU-2015.5>. Accessed April 1, 2024.
- Fasseeh A, Karam R, Jameleddine M, et al. Implementation of health technology assessment in the Middle East and North Africa: comparison between the current and preferred status. *Front Pharmacol*. 2020;11:15.
- Kennedy-Martin M, Slaap B, Herdman M, et al. Which multi-attribute utility instruments are recommended for use in cost-utility analysis? A review of national health technology assessment (HTA) guidelines. *Eur J Health Econ*. 2020;21(18):1245–1257.
- Rowen D, Azzabi Zouraq I, Chevrou-Severac H, van Hout B. International regulations and recommendations for utility data for health technology assessment. *Pharmacoeconomics*. 2017;35(suppl 1):11–19.
- Shaw C, et al. A review of the use of EQ-5D for clinical outcome assessment in health technology assessment, regulatory claims, and published literature. *Patient Educ Couns*. 2024;17(3):239–249.
- Ahmad M, Akhras KS, Saleh S. Genuine policy learning is fundamental: the journey of the United Arab Emirates toward the establishment of health technology assessment. *Int J Technol Assess Health Care*. 2023;39(1):e3.
- AlAujan SS, Almazrou SH, Al-Aqeel SA. A systematic review of sources of outcomes and cost data utilized in economic evaluation research conducted in the Gulf Cooperation Council. *Risk Manag Healthc Policy*. 2021;14:209–220. ISPOR. Pricing and reimbursement: issues and challenges. <https://www.ispor.org/member-groups/global-groups/networks/arabic-network/ispor-dubai-2018/>; Published 2018. Accessed April 1, 2024.
- Koornneef E, Robben P, Blair I. Progress and outcomes of health systems reform in the United Arab Emirates: a systematic review. *BMC Health Serv Res*. 2017;17(1):672.
- Nagi MA, PEN Dewi, Thavorncharoensap M, Sangroongruangsri S. A systematic review on economic evaluation studies of diagnostic and therapeutic interventions in the Middle East and North Africa. *Appl Health Econ Health Policy*. 2022;20(3):315–335.
- Nagi MA, Luangsinsiri C, Thavorncharoensap M. A systematic review of economic evaluations of vaccines in Middle East and North Africa countries: is existing evidence good enough to support policy decision-making? *Expert Rev Pharmacoecon Outcomes Res*. 2021;21(16):1159–1178.
- Dubai Go. Dubai Health Authority Introduces Value-Based Healthcare for the Emirate; 2022.
- Papadimitropoulos EA, Elbarazi I, Blair I, Katsaiti MS, Shah KK, Devlin NJ. An investigation of the feasibility and cultural appropriateness of stated preference methods to generate health state values in the United Arab Emirates. *Value Health Reg Issues*. 2015;7:34–41.
- Global Media Insight. United Arab Emirates (UAE) population statistics. 2024.